

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

REGISTRATION
NUMBER

BIOLOGY

9744/03

Paper 3: Long Structured and Free-response Questions

26 August 2022

Candidates answer on the Question Paper.

2 hours

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and Biology class in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your workings or if you do not use appropriate units.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/30
2	/10
3	/10
Section B	
4/5	/25
Total	/75

This document consists of **23** printed pages and **1** blank page.

Section A

Answer **all** the questions in this section.

- 1 The influenza A virus subtype H7N9 has caused several epidemic waves since its appearance in 2013. H7N9 influenza virus causes high-mortality disease in humans. Monoclonal antibodies, made by cloning a unique white blood cell are being explored as treatments.

Fig. 1.1 shows how one such antibody, m826 antibody, could bind to hemagglutinin (HA) of the H7N9 influenza virus.

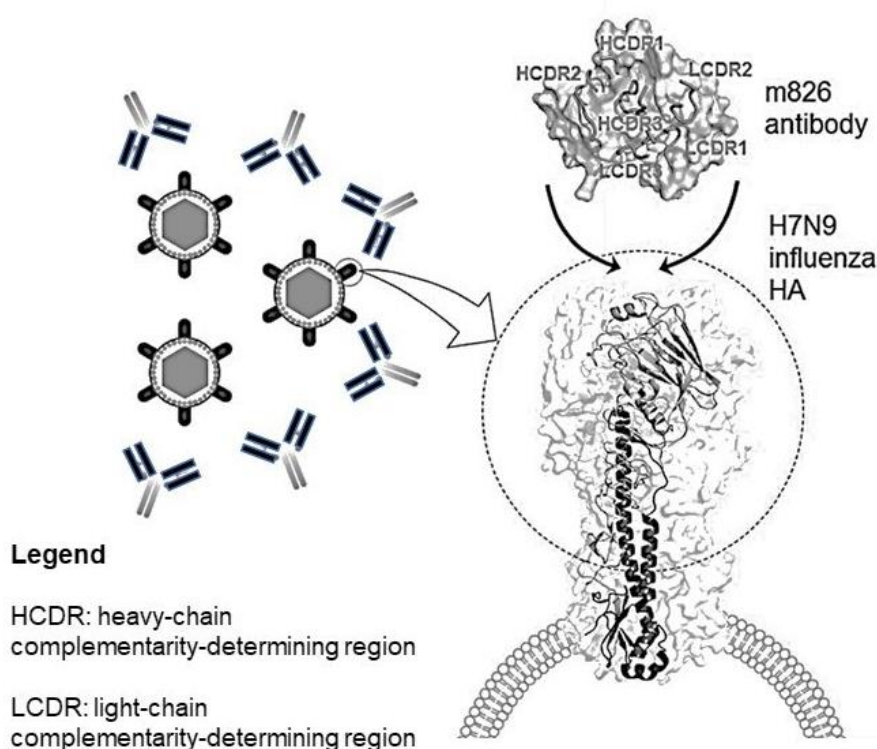


Fig. 1.1

- (a) Name and describe the secondary structures of HA that are visible in Fig. 1.1.

LO 1(m); AOB

Answers:

1. α -helices/helix;
2. β -sheet(s);
3. C=O group of one amino acid residue is hydrogen bonded to the N-H group;
4. (α -helices) ... four amino acid residues further up the chain / (β -sheets) ... of another amino acid residue on a neighbouring chain;

[4]

- (b) With reference to Fig. 1.1, describe how m826 antibody could bind to HA of the H7N9 influenza virus.

LO 1(m), A(c); AOB

Answers:

1. ref to presence of HCDRs and LCDRs of antigen-binding site of m826 antibody;
2. ref to antigen-binding site of m826 antibody being complementary in shape and charge to the exposed site on HA;
3. ref to formation of hydrogen bonds / ionic bonds / hydrophobic interactions / van der Waals forces (name at least two types of bonds) between amino acid residue(s) of m826 antibody and that of HA;

R: formation of disulfide bonds as covalent bonds never occur between antigens and naturally produced antibodies

R: active site

[3]

- (c) Membrane glycoproteins such as HA consist of one or more short carbohydrate chains attached to the protein.

Outline the functions of such short carbohydrate chains on the cell surface membrane of the human epithelial cell.

LO 1(k); AOA

Answers:

1. ref to cell-cell recognition (A: cell recognition)
2. ref to cell-cell adhesion / cell contact cell to form tissue (A: cell adhesion)
3. receptor-mediated endocytosis
4. ligand-receptor interaction (R: cell signaling, vague)

R: substrate recognition, cell communication

Note that such carbohydrate chains are present (as part of glycoproteins / glycolipids) only on the outer leaflet of the cell surface membrane.

[2]

- (d) HA is embedded in the influenza viral envelope that is derived from the cell surface membrane of a human epithelial cell of the respiratory tract.

State **two** structural similarities between the influenza viral envelope and the cell surface membrane of the human epithelial cell.

LO 1(e), 1(j); AOB

Answers:

1. consists of a phospholipid bilayer;
2. contains (glyco)proteins / membrane protein
3. contains cholesterol;

[2]

- (e) The fluid mosaic model of the cell surface membrane was elucidated with the aid of freeze-fracture electron microscopy. The steps involve freezing and fracturing the membrane before examining it with a scanning electron microscope, which produces images of an object's surface.

Fig. 1.2 shows the process of freeze-fracture electron microscopy.

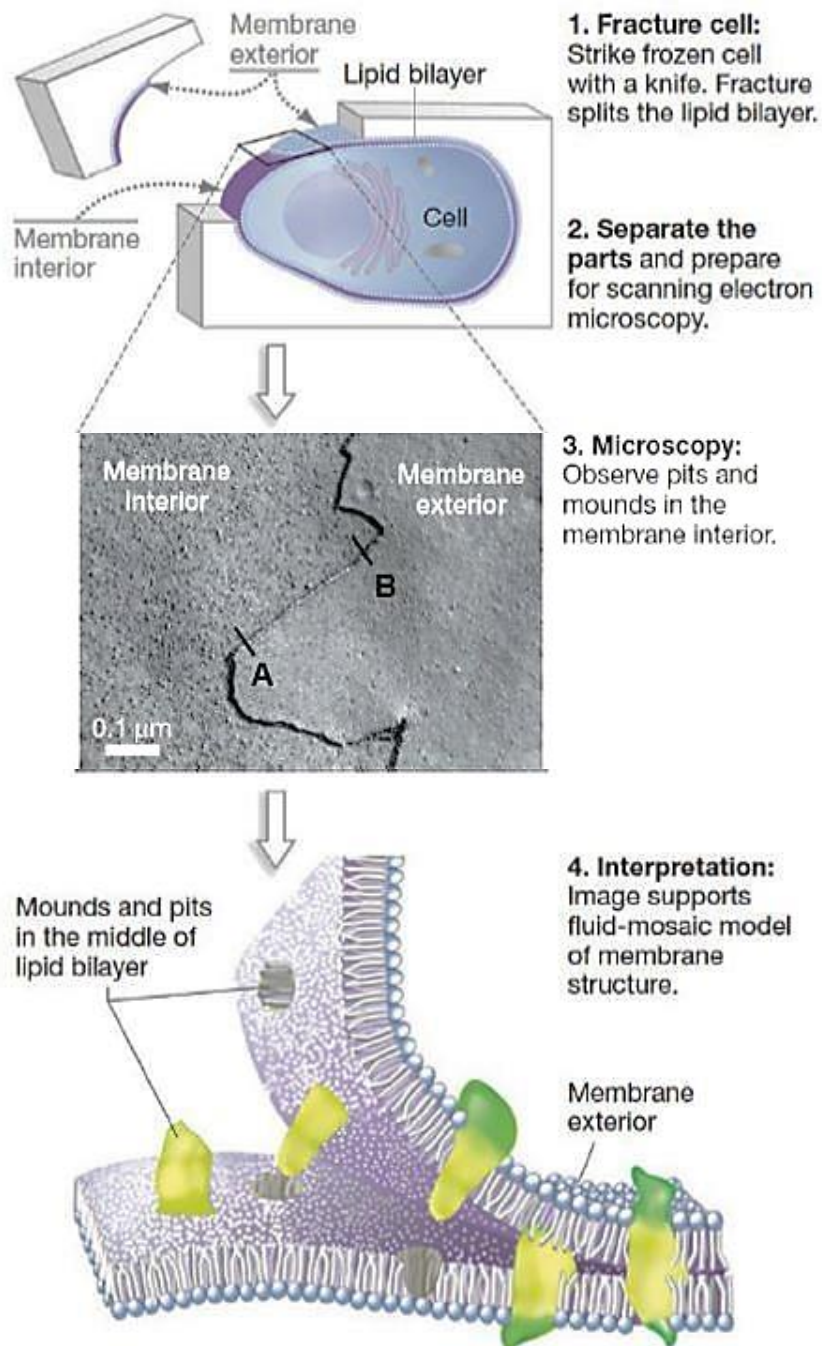


Fig. 1.2

- (i) Determine the distance between points **A** and **B** as shown in Fig. 1.2.

Show your working clearly in the space provided.

POS; AOB

Answers:

1. distance between points **A** and **B** = 19/20 mm or 1.9/2.0 cm
2. Calculation: example, $2/0.7 \times 0.1$
3. Final answer on the line = 0.286 μm (2 or 3 s.f.)

distance between points **A** and **B** = _____ μm [3]

Other scenarios:

If 0.1 μm = 0.8 cm

2 cm $\rightarrow$ 0.25 μm

1.9 cm $\rightarrow$ 0.238 μm

If 0.1 μm = 0.75 cm

2 cm = 0.2666 μm

1.9 cm = 0.2533 μm

- (ii) Suggest an appropriate control to show that the mounds and pits were not irregularities in the lipid bilayer caused by the freeze-fracture process.

POS; AOB

Answers:

Repeat the freeze-fracture process using a lipid bilayer / liposome / vesicle that consists of only phospholipid / no proteins embedded;

[1]

- (f) Class switching results in millions of different antibody molecules. During class switching, the variable region generated during V(D)J recombination remains unchanged, but it becomes linked to another segment of the constant region of the heavy chain.

Fig. 1.3 shows how human genomic IgM-IgD region may undergo transcription in a mature B lymphocyte to produce mRNA coding for secreted or transmembrane form of antibody molecules (IgM or IgD). Exons encoding the rearranged VDJ region and the various domains of IgM and IgD are shown in boxes.

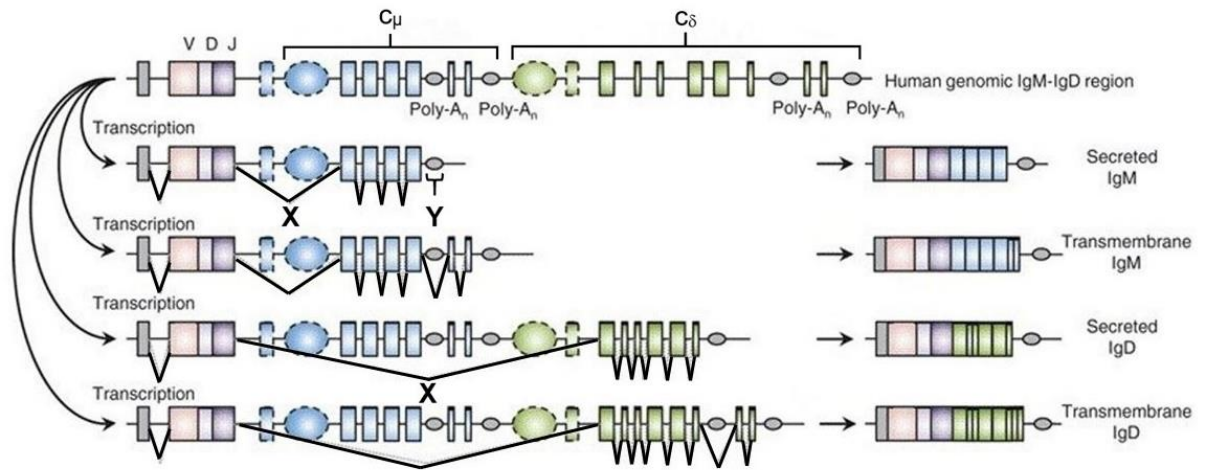


Fig. 1.3

- (i) State the **level** of control of gene expression as shown in Fig. 1.3.

LO 2(j)(iii); AOB

[1]

Answers:

post-transcriptional; (reject: splicing)

- (ii) Name processes X in Fig. 1.3 and describe their significance.

LO 2(j)(iii); AOB

Answers:

1. alternative splicing; (reject: splicing, no ecf)
2. enables an organism to produce more proteins while storing a compact genome / enables different proteins to be produced from a single gene / one heavy-chain gene can code for two different classes (IgM and IgD) of antibody molecules with different effector functions;

[2]

- (iii) With reference to Fig. 1.3, identify region Y on the mRNA coding for secreted IgM and describe its roles.

LO 2(j)(iii); AOB

Answers:

1. region Y = poly-A signal sequence / polyadenylation signal sequence; R: poly-A tail

Any two from below

2. ref to binding of an enzyme (poly-A polymerase) to region Y;
3. ref to protection of mRNA from rapid degradation by 3' exonucleases / ribonucleases ; R: endonucleases/nucleases
4. ref to facilitation of mRNA transport across the nuclear membrane;

[3]

- (iv) On Fig. 1.3, circle the region of the mRNA that codes for the transmembrane region of IgD.

LO 2(c); AOB

[1]

Answers:



- (v) Explain the structure and function of the transmembrane region of IgD.

LO 1(j); AOA

Answers:

1. Consists of hydrophilic amino acids / Stretches of amino acids with polar or charged side chains
 2. interact with the phosphate head of the phospholipid bilayer
- OR
3. consists of hydrophobic amino acids / Stretches of amino acids with nonpolar side chains
 4. interact with the fatty acids in the interior of the phospholipid bilayer, away from water;

- (g) Fig. 1.4 shows the excess deaths caused by pneumonia and influenza over a 50-year period in Country P and Country Q. The five-year moving average is also shown. The shaded bars show the rates of use of vaccine in each country.

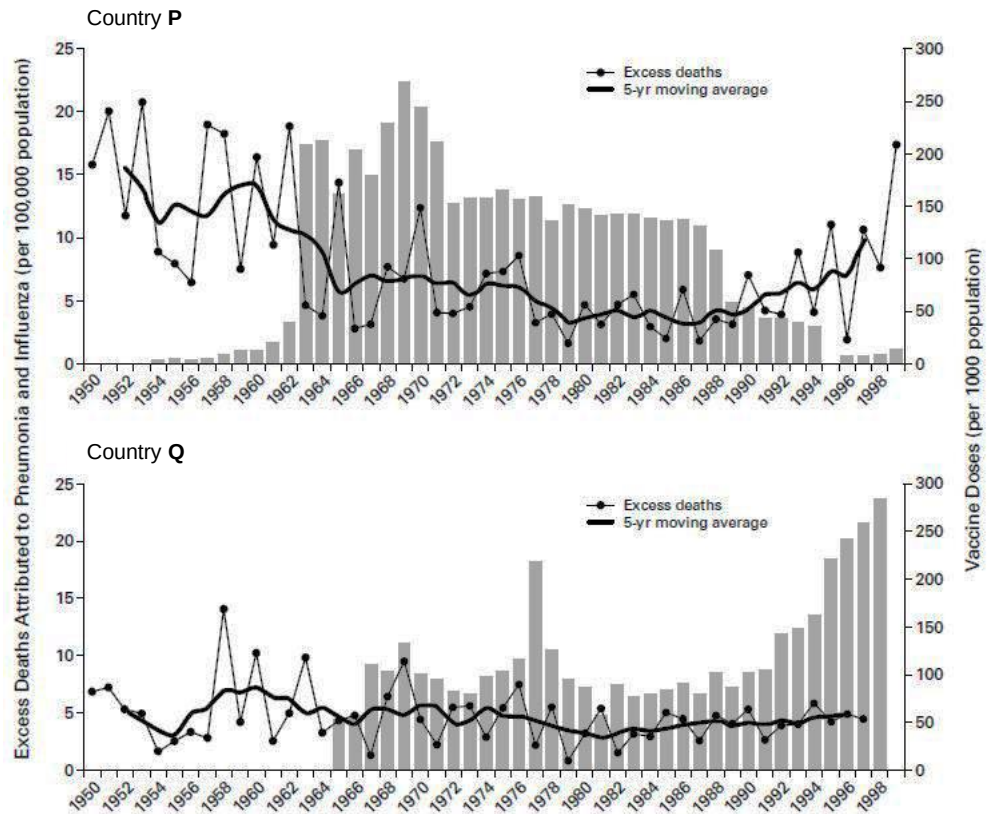


Fig. 1.4

- (i) With reference to Fig. 1.4, describe the effect of vaccination on excess deaths caused by pneumonia and influenza in the two countries.

AOB

Answers:

[Maximum 2 marks for any describe+quote pair for Country P]

Pd1.

In country P, as the rates of use of vaccine / number of vaccine doses (per 1000 population) per year increased, the number of excess deaths attributed to pneumonia and influenza (per 100,000 population) decreased;

Pq1.

In country P, as the rates of use of vaccine / number of vaccine doses (per 1000 population) per year increased from 0 in 1953 to 264 (accept 260-270) in 1969, the number of excess deaths attributed to pneumonia and influenza (per 100,000 population) decreased from 21 in 1953 to 7 in 1969;

Pd2.

In country **P**, as the rates of use of vaccine / number of vaccine doses (per 1000 population) per year decreased, the number of excess deaths attributed to pneumonia and influenza (per 100,000 population) increased;

Pq2.

In country **P**, as the rates of use of vaccine / number of vaccine doses (per 1000 population) per year decreased from 150 in 1979 to 14 (accept 10-20) in 1999, the number of excess deaths attributed to pneumonia and influenza (per 100,000 population) increased from 2 in 1979 to 17-18 in 1999;

Pd3.

In country **P**, as the rates of use of vaccine / number of vaccine doses (per 1000 population) per year remained (relatively) high, the number of excess deaths attributed to pneumonia and influenza (per 100,000 population) remained (relatively) low / decreased slightly;

Pq3.

In country **P**, as the rates of use of vaccine / number of vaccine doses (per 1000 population) per year remained (relatively) high at around 150 from 1972 to 1987, the number of excess deaths attributed to pneumonia and influenza (per 100,000 population) remained (relatively) low at around 2-4 from 1972 to 1987/ decreased slightly from 4 in 1972 to 2 in 1987;

[Maximum 2 marks for any quote+describe pair for Country Q]

Qd1.

In country **Q**, as the rates of use of vaccine / number of vaccine doses (per 1000 population) per year increased, the number of excess deaths attributed to pneumonia and influenza (per 100,000 population) remained (relatively) constant;

Qq1.

In country **Q**, as the rates of use of vaccine / number of vaccine doses (per 1000 population) per year increased from 100 in 1990 to 258 (accept 255-260) in 1997, the number of excess deaths attributed to pneumonia and influenza (per 100,000 population) remained (relatively) constant at around 4-5 from 1990 to 1997;

Qd2.

In country **Q**, as the rates of use of vaccine / number of vaccine doses (per 1000 population) per year remained (relatively) low, the number of excess deaths attributed to pneumonia and influenza (per 100,000 population) showed greater deviation from the 5-yr moving average;

Qq2.

In country **Q**, as the rates of use of vaccine / number of vaccine doses (per 1000 population) per year remained (relatively) low at around 100 from 1967 to 1982, the number of excess deaths attributed to pneumonia and influenza (per 100,000 population) showed greater deviation as much as +/- 5 from the 5-yr moving average from 1967 to 1982;

[4]

- (ii) The policies for the control of pneumonia and influenza differed between Country **P** and Country **Q**. In Country **P**, vaccination was recommended for school children while in Country **Q**, vaccination was recommended for the elderly.

Using this information, suggest **one** reason for the trends described in (g)(i).

LO A(e), A(f); AOB

[2]

[Total: 30]

	Reason	Explanation
1	Mobility	Children more mobile interact with more people. Elderly, less mobile.
2	Exposure to crowd	Children higher exposure to crowd OR elderly less exposure to crowd
3	Attribution of access to other age groups	e.g. in country P, children are small group so fluctuations due to other age groups.
4	Attribution to waning of immune response	Excess deaths due to loss of immunological memory in other age groups
5	Attribution to existent adaptive immunity	In P, children has no adaptive immunity, vaccination increases immunity so average death decreases. In Q, most elderly has some adaptive immunity. Therefore vaccination did not further decrease average death
6	Attribution to larger demographic of young in P	When vaccination increase, average excess death decreases
7	Attribution to vaccination more effective for young	Excess death decrease by a greater extent in P than in Q.
8		

- 2 An increase in atmospheric carbon dioxide (CO₂) concentration and temperature affect the growth and yield of important crops such as potatoes.

In 2020, researchers investigated the effect of long-term exposure to elevated CO₂ concentration and temperature on potato plants.

The treatments involved the conditions: ambient CO₂ concentration and ambient temperature (aCaT), ambient CO₂ concentration and elevated temperature (aCeT), elevated CO₂ concentration and ambient temperature (eCaT), and concurrently elevated CO₂ concentration and temperature (eCeT). Conditions were maintained using a growth chamber for each of the treatments.

The ambient temperature was maintained at 20°C as modern potato cultivars generally grow well under moderate temperatures around 20°C while the elevated temperature was maintained at 4°C higher than ambient temperature.

Table 2.1 shows the details of the experiment conducted.

Table 2.1

	aCaT	eCaT	aCeT	eCeT
daily mean temperature / °C	20.9	20.9	24.9	24.9
daily mean CO ₂ concentration / ppm	417	771	425	804

- (a) Suggest why aCaT and aCeT use a CO₂ concentration at approximately 400 ppm.

LO: POS

Answers:

Current concentration of CO₂ in the atmosphere.

[1]

Growth parameters such as plant height, leaf area and biomass were measured for six plants from each chamber at 58 days after emergence. Emergence is the growth of the plant towards the soil surface and coming out from the soil, making a shoot.

Fig. 2.1 shows the effect of different treatments on growth parameters at 58 days after emergence. Error bars are the lines above the vertical bars. Different letters above the error bars indicate significant differences between the treatments while same letters indicate no significant differences between the treatments.

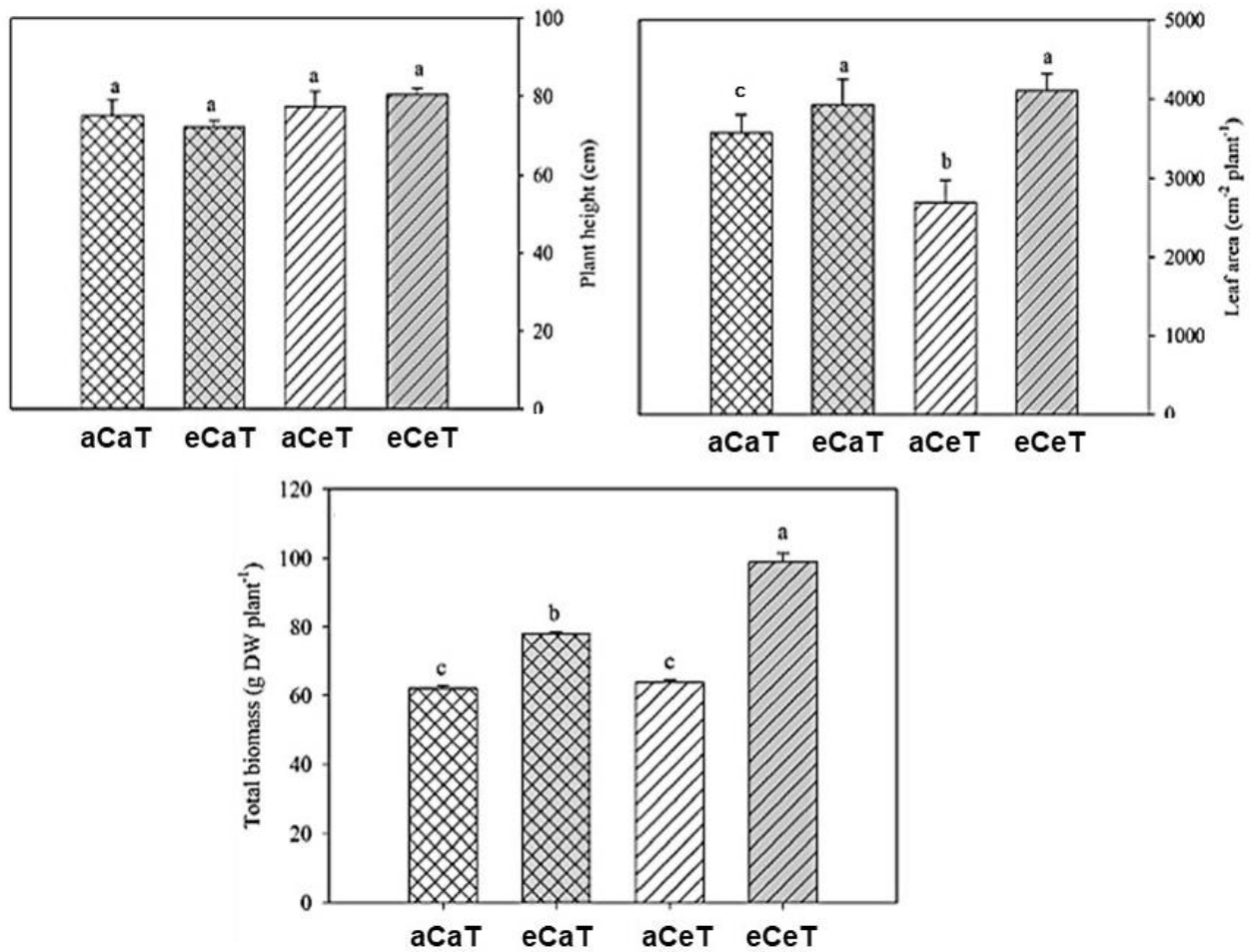


Fig. 2.1

(b) Complete Table 2.2 to show the conclusions that could be drawn from Fig. 2.1.

Table 2.2

parameter	conclusion	
	elevated CO ₂ concentration	elevated temperature
plant height	no significant differences	No significant differences
leaf area	Significant increase	significant decrease
total biomass	Significant increase	No significant differences

[4]

- (c) With reference to Fig. 2.1 and your knowledge on photosynthesis, explain the results for total biomass in the treatment eCaT.

LO: 3(c) POS

Answers:

1. Total biomass of control (aCaT) at 61 g DW plant⁻¹ whereas total biomass for eCaT treatment increased to 79/80 g DW plant⁻¹

Any 1 point below:

2. Ref: increase CO₂ concentration → increase rate of effective collision with rubisco
3. Ref: increase rate of carbon fixation
4. Ref: increase rate of G3P formed → use for growth

[2]

At harvest time, the average numbers of potato tubers and biomass of potatoes cultivated under varying CO₂ concentrations and temperatures were collected.

Table 2.3 shows the results.

Table 2.3

treatment	average numbers of potato tubers / plant ⁻¹			biomass / g DW plant ⁻¹
	< 30 g	30 – 80 g	> 80 g	
aCaT	4.8	2.9	2.3	106.4
eCaT	5.2	3.9	2.4	110.3
aCeT	2.0	2.8	1.9	76.9
eCeT	3.9	4.1	2.9	127.5

- (d) With reference to Table 2.3, describe the effect of elevated temperature on biomass of potato tubers at harvest time.

LO: POS

Answers:

1. Decrease in biomass for potatoes grown in elevated temperature (76.9 g DW plant⁻¹) compared to ambient temperature (106.4 g DW plant⁻¹)
2. Increase in biomass for potatoes grown in eCeT (127.5 g DW plant⁻¹) compared to ambient temperature eCaT (106.4 g DW plant⁻¹)

[1]

- (e) “Global warming due to increased atmospheric CO₂ concentration will lead to reduced global food supply”.

Discuss the validity of this statement.

LO: POS

Answers:

1. Statement is not valid
2. Ref: increased CO₂ concentration will lead to increase in Biomass although temperature increased.

If point 1 is wrong, do not mark for point 2.

[2]

[Total: 10]

- 3 The extreme coat-colour variations in oldfield mouse, *Peromyscus polionotus*, are driven by selection for camouflage. The mainland subspecies, *P. p. subgriseus*, has a coat that is dark brown on top and light grey on the belly, as well as a striped tail. The beach subspecies, *P. p. leucophalus*, has a light-coloured coat and lacks visible pigmentation on its face, sides and tail.

Fig. 3.1 shows the pigmentation of the two subspecies.

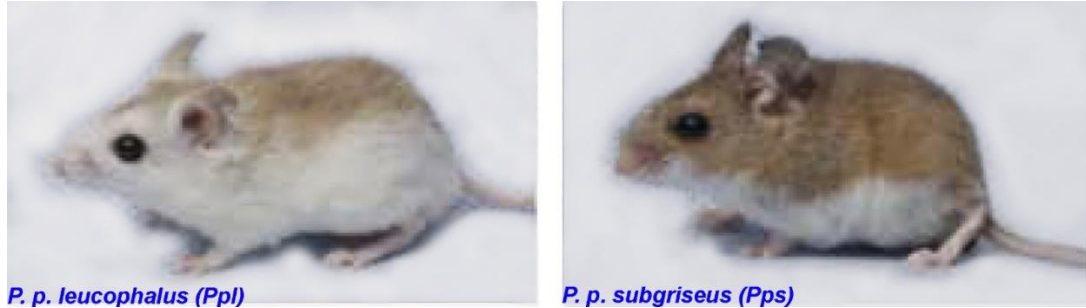


Fig. 3.1

Two genes, *melanocortin-1 receptor (Mc1r)* and its antagonist, *Agouti signaling protein (Agouti)*, located on independent chromosomes are responsible for most of the differences in pigmentation between the two subspecies. A receptor antagonist is a type of ligand that blocks or dampens a biological response when bound to the receptor.

To examine whether differences in the expression level of *Mc1r* or *Agouti* influence colour patterning, reverse transcriptase polymerase chain reaction [RT-PCR] was carried out on adult skin cells of the eyebrow area. The resultant RT-PCR product was analysed by gel electrophoresis.

Fig. 3.2 shows the results of the gel electrophoresis.

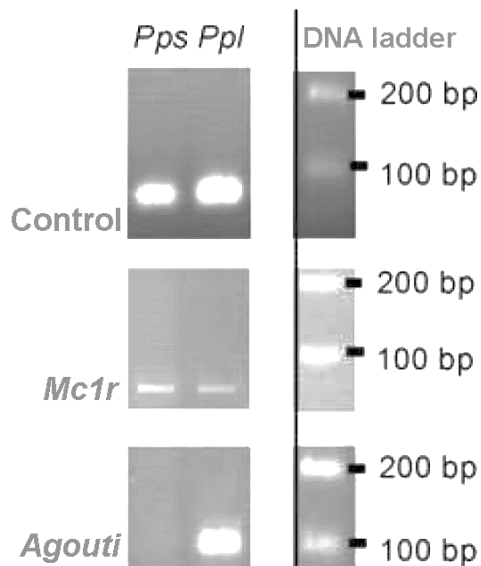


Fig. 3.2

(a) With reference to Fig. 3.2,

(i) describe the principle of gel electrophoresis.

LO 2(k); AOA

Answers:

1. separates DNA fragments by size / length; R: molecular weight/mass

Any 1 below:

2. smaller DNA fragments move faster towards positively-charged pole + lower position of 100bp band compared with 200bp band;

3. similar/ same sized DNA fragments move at the same speed + Mc1r bands in both Pps and Ppl are the same size/ Agouti fragment is approximately 100bp;

4. thickness/ brightness of band reflects relative quantity of DNA + more control DNA in Ppl than Pps/ approximately same amount of Mc1r in both Pps and Ppl;

[2]

(ii) describe the expression level of *Mc1r* and *Agouti*.

LO 2(j); AOB

Answers:

1. *Mc1r* is expressed in both subspecies;

2. *Agouti* is expressed only in beach subspecies / ppl ;

[2]

(b) RT-PCR is a laboratory technique combining reverse transcription of RNA into DNA and amplification of specific DNA targets using polymerase chain reaction. The technique is carried out on mRNA samples isolated from cells.

Suggest **one** advantage of RT-PCR compared to conventional PCR.

LO 2(k); AOB

Answers:

1. RT-PCR can provide information on **spatial expression patterns**

2. RT-PCR can provide information on **temporal gene expression patterns**, as only genes are expressed as mRNA will be amplified;
3. Can show level of expression

[1]

To analyse the genetic basis of pigmentation pattern differences, mainland mice were crossed with beach mice, yielding F_1 hybrids that were crossed with each other to produce F_2 progeny.

(c) Alleles for *Mc1r* and *Agouti* are represented by **D** for dark and **L** for light.

Table 3.1 shows the distribution of genotypes in 30 F_2 progeny lacking pigmentation on the rear end.

Table 3.1			
	DD	DL	LL
<i>Mc1r</i>	0	0	30
<i>Agouti</i>	0	15	15

State the dominant allele for each of the following genes.

Mc1r dark allele/ D allele;

Agouti light allele/ L allele;

[2]

Reject: d and l

- (d) Fig. 3.3 shows the pigmentation in hairs for pigmented cheek area in 454 F₂ progeny clustered by *Agouti* genotype.

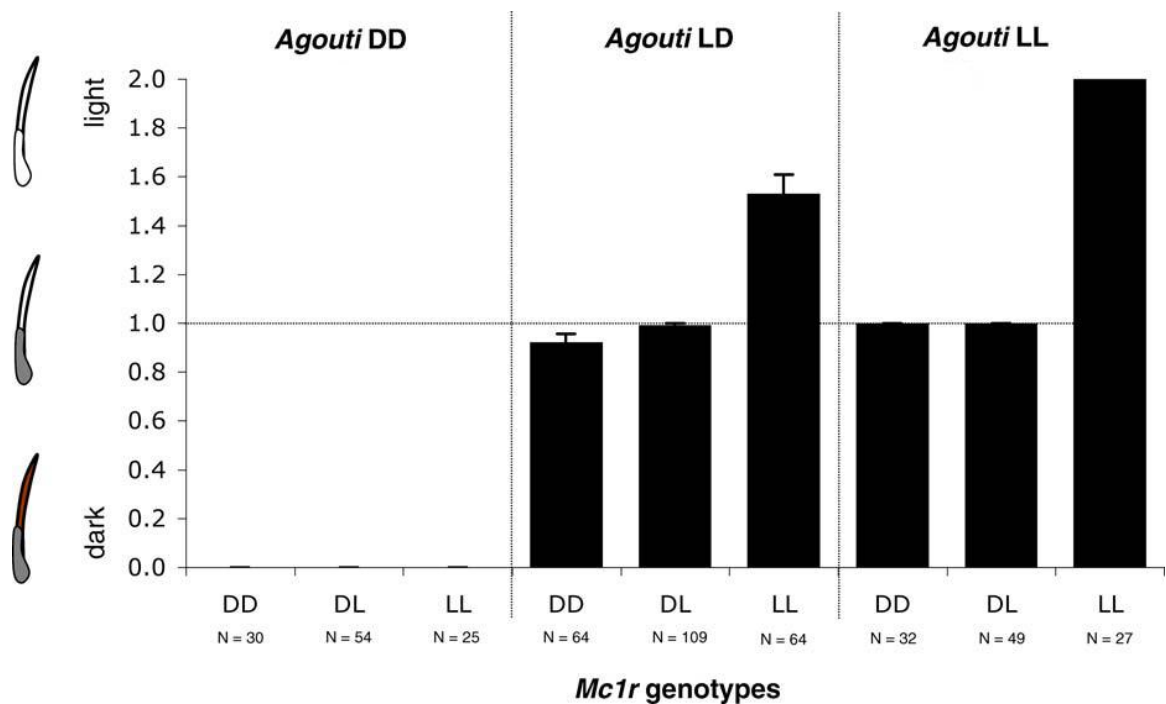


Fig. 3.3

The phenotype for each of 454 F₂ progeny was scored as 2 (no visible pigment), 1 (partially pigmented), and 0 (fully pigmented) as indicated by the cartoons of individual hairs. N refers to the number of individuals in each category.

Name and describe the type of gene interaction as shown in Fig. 3.3.

LO 2(aa); AOA

Answers:

1. epistasis;

Any 2 below:

2. gene at one locus alters the phenotypic expression of a gene at a second locus;
3. Recessive epistasis/ *Agouti* DD masks effects of both alleles of *Mc1r*;
4. *Agouti* DD is epistatic to *Mc1r* gene
5. *Mc1r* genotype is expressed fully in *Agouti* LL;
6. *Mc1r* genotype is not fully expressed in presence of one *Agouti* D allele;

[3]

[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

4 (a) Describe how a functional enzyme is synthesised in bacteria. [15]

A. Describe transcription: (max 5m)

- A1 Ref. to transcription of gene on template strand
- A2 Ref. to RNA polymerase binding to promoter region / pribnow box
- A3 Ref. to RNA polymerase adding free ribonucleoside triphosphates / ribonucleotides and are joined together by phosphodiester bonds
- A4 Ref. to elongation of mRNA in a 5' to 3' direction
- A5 Ref. to DNA read in a 3' to 5' direction
- A6 Ref. to transcription stops after RNA polymerase reads the termination signal (in prokaryotes) / polyadenylation signal (in eukaryotes)

B. Describe translation: (max 5m)

- B1 Translation can begin immediately once mRNA is synthesized by RNA polymerase /
- B2 Ref. to initiation of translation at 5' end of mRNA by assembling small and large ribosome and initiator tRNA / ref: tRNA for methionine;
- B3 Ref. to tRNA with complementary anticodon forms complementary base pairs with codon sequences, bringing specific amino acids to ribosome;
- B4 Ref. to formation of peptide bonds catalysed by peptidyltransferase in large ribosomal subunit;
- B5 Ref to stop codon and addition of release factor to A site;
- B6 Ref. to hydrolysis of ester bond between amino acid and tRNA, releasing polypeptide chain from ribosome;
- B7 Formation of polyribosome
- B8 Ribosome moves in a 5' to 3' direction

C. Describe protein folding: (max 5m)

- C1 Enzymes are globular proteins with unique three-dimensional conformation / tertiary / quaternary structure;
- C2 Ref to primary structure being the unique sequence and number of amino acids in a polypeptide linked by peptide bonds;
- C3 Tertiary structure refers to the folding of polypeptide into a specific conformation, held by bonds between R-groups* of structural amino acids within same polypeptide, maintained by hydrophobic interaction, hydrogen bonds, ionic bonds, disulfide bridges;
- C4 Ref to quaternary structure: more than 1 polypeptide chain to form functional protein held by hydrophobic interaction, hydrogen bonds, ionic bonds, disulfide bridges between R groups between polypeptide chains;
- C5 Folding gives rise to a specific cleft / groove - active site that is complementary in shape and charge to its substrate.
- C6 Folding brings catalytic amino acids far apart in the primary structure / polypeptide close together in the active site
- C7 Amino acid residues at the active site create a suitable microenvironment

Overall max 15m. QWC min. 1 point from segment A, B and C

(b) Outline the process of transduction and its advantages to the bacteria.

[10]

A. Transduction Process

- A1 Transduction is the process by which bacterial DNA / genes is transferred from one bacterium (host cell) to another (recipient cell) via a bacteriophage.
- A2 Generalised transduction requires infection of a bacterium by a virulent/T4 bacteriophage R: lytic phage
- A3 During the assembly of the newly replicated phage genome within the phage capsid, a small piece of the host cell's DNA gets mistakenly packaged within the capsid / the head (P: new virus)
- A4 Specialised transduction requires infection of a bacterium by a temperate/lambda bacteriophage R: lysogenic phage
- A5 When a temperate phage enters lytic cycle from lysogenic cycle / spontaneous induction occurs
- A6 Small region of the bacterial DNA that was adjacent to the prophage is excised;
- A7 *Mark point deleted*
- A8 **Ref:** The defective phages infect / attach to other bacterial cells and inject the piece of host bacterial DNA into the newly infected bacterial cell cytoplasm;

A9 Ref to: homologous recombination / crossing over

(Generalised & Specialised transduction) Foreign bacterial DNA is incorporated into the recipient / bacterial cell's chromosome through homologous recombination / If there is sufficient homology between the DNA fragments and bacterial chromosome, crossing over occurs, and segments of the original chromosomal DNA will be replaced;

AVP e.g. Hydrolysis of bacteria genome for lytic cycle

B. Advantages

B1 New alleles can then be incorporated into the bacterial genome of recipient cell thus recipient cell will subsequently express new characteristics / show change in phenotype;

B2 Generate genetic diversity / variation;

B3 (Example of new allele) Gain antibiotic resistant alleles from other bacteria;

B4 (How it confers selective advantage) Antibiotics less effective against the bacteria;

B5 Thus the bacteria is at a selective advantage / **higher fitness**

QWC:

At least one point from A and one point from B

[Total: 25]

5 (a) Describe the structure-function relationship of the endomembrane system in plants. [15]

A. Introduction on the endomembrane system: (max 1m)

A1 The endomembrane system encompasses the nuclear envelope, the endoplasmic reticulum, the Golgi body, lysosomes, various kinds of vesicles and vacuoles, and the cell surface membrane (name at least three components of the endomembrane system);

A2 The membranes of this system are related either through direct physical continuity or by the transfer of membrane segments as tiny vesicles;

B. Nuclear envelope: (max 2m, i.e. 1 pair of s+f)

B1s Ref. to the presence of nuclear pores in nuclear envelope;

B1f1 Ref. to the exchange of substances between the nucleus and cytoplasm;

B1f2 Ref. to at least two examples (mRNA / tRNA / ribosomal subunits) of substances that exit the nucleus via nuclear pores;

B1f3 Ref. to at least two examples (nucleotides / ribosomal proteins / hormones for regulation of gene expression) of substances that enter the nucleus via nuclear pores;

C. Endoplasmic reticulum: (max 6m, i.e. 3 pairs of s+f)

C1s Ref. to the presence of ribosomes (not permanently) bound to (the cytosolic side of) the rough endoplasmic reticulum (ER) membrane;

C1f Ref. to the rough ER as the site of synthesis of secretory proteins / (trans)membrane proteins;

C2s Ref. to the presence of enzymes embedded in the rough ER membrane;

C2f Ref. to the rough ER as the site of assembly of phospholipids from precursors in the cytosol / site of chemical modifications of secretory proteins / site of synthesis of glycoproteins;

C3s Ref. to the ER membrane separating the ER lumen / cisternal space from the cytosol;

C3f Ref. to the ER lumen / cisternal space as the site of folding and assembly of the newly synthesised water-soluble polypeptide chains;

C4s Ref. to the presence of enzymes embedded in the smooth ER membrane;

C4f Ref. to the smooth ER as the site of synthesis of lipids;

D. Golgi body: (6m, i.e. 3 pairs of s+f)

D1s Ref. to the Golgi body consisting of stacks of cisternae / curved flattened membranous sacs that have different assortments of enzymes anchored to the membrane;

- D1f Ref. to the progressive processing (e.g. dehydration of proteins / emulsification of lipids, glycosylation and modifications of the carbohydrate chains of glycoproteins) of products (proteins and phospholipids) of the ER as they travel from the cisternae to the *trans* face of the Golgi body / the Golgi body as the site of synthesis and secretion of extracellular polysaccharides (e.g. pectins) that are used as components of the plant cell wall;
- D2s Ref. to the presence of the *cis* face of the Golgi body that is located nearest to the nucleus and ER;
- D2f Ref. to the *cis* face receiving materials from transport vesicles budding from the ER;
- D3s Ref. to the presence of the *trans* face of the Golgi body that is located nearest to the cell surface membrane;
- D3f Ref. to the *trans* face packaging molecules in vesicles and transporting them out of the Golgi body;

E. Vacuole(s): (max 2m, i.e. 1 pair of s+f) (reject: lysosomes)

- E1s Ref. to the presence of cell sap / water / fluid in the large central vacuole;
- E1f Ref. to the maintenance of turgor pressure in a plant cell by the large central vacuole;
- E2s Ref. to the presence of hydrolytic enzymes in the vacuole(s);
- E2f Ref. to the vacuole(s) as the site of digestion of various macromolecules (e.g. proteins, nucleic acids, polysaccharides);
- E3s Ref. to the acidic environment of about pH 5 found in vacuole(s);
- E3f Ref. to the activity of the hydrolytic enzymes being optimum only in vacuole(s) / not very active in the cytosol of about pH 7.4;

F. Cell surface membrane: (2m, i.e. 1 pair of s+f)

- F1s Ref. to the cell surface membrane consisting of a phospholipid bilayer / the fluid mosaic model of the cell surface membrane / the presence of cellulose synthase embedded in the cell surface membrane;
- F1f Ref. to the cell surface membrane as the site of membrane fusion / exocytosis / endocytosis / cellulose synthesis;

QWC (compulsory mark):

At least one pair of s+f described for at least three different components of the endomembrane system;

(b) Outline the process of photophosphorylation and its advantages.

[10]

A. Process: (max 7m)

- A1 Ref. to the thylakoid membrane as the site of (non-cyclic / cyclic) photophosphorylation in chloroplasts;
- A2 Ref. to the presence of photosystems / electron transport chains (ETC) / series of electron carriers / ATP synthase found on the thylakoid membrane;
- A3 Ref. to the passing of photoexcited electrons from the primary electron acceptor of photosystem II to the first ETC / from the primary electron acceptor of photosystem I to the second ETC;
- A4 Ref. to the transport of electrons down the first ETC until they reach P700⁺ of photosystem I / down the second ETC until they reach NADP⁺ (the final electron acceptor);
- A5 Ref. to the cyclic electron flow from ferredoxin to the cytochrome complex and from there continue on to P700⁺ of photosystem I;
- A6 Ref. to the energy released from the electron transport being used to pump protons / hydrogen ions / H⁺ across the thylakoid membrane from the stroma into the thylakoid space of the chloroplast;
- A7 Ref. to the proton reservoir formed in the thylakoid space contributing to ATP synthesis via chemiosmosis;
- A8 Ref. to ATP being synthesised by ATP synthase from ADP and phosphate;

B. Advantages: (min 2m)

- B1 Ref. to photophosphorylation allowing the conversion of solar/light energy into chemical energy;
- B2 Ref. to photophosphorylation producing ATP for light-independent reactions / Calvin cycle of photosynthesis;
- B3 Ref. to photophosphorylation producing oxygen (gas) for cellular/aerobic respiration / to sustain life on Earth;

QWC (compulsory mark):

At least one point from A and one point from B;

